MTH101, Basic Linear Algebra

Monsoon 2025

30/10/2025, 2:00PM-3:00PM.

Tutorial-6

(1) Let W_1 and W_2 be two subspaces of V. Prove that $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

(2) Let W_1 and W_2 be two subspaces of V. Prove the following statements.
   (a) $W_1 + W_2 := \{w_1 + w_2 : w_1 \in W_1, w_2 \in W_2\}$ is a subspace of V.
   (b) The smallest subspace containing $W_1 \cup W_2$ is the subspace $W_1 + W_2$.

(3) Prove that any three vectors in $\mathbb{R}^2$ are linearly dependent.

(4) Let $v_1, \ldots, v_n$ be linearly independent vectors in V and $v \in V$. Prove that $v, v_1, \ldots, v_n$ are linearly independent if and only if $v \notin \mathrm{span}\{v_1, v_2, \ldots, v_n\}$.